Algorithmiq 竞争对手档案—详细数据清单

生成日期: 2026-06-17 数据来源: algorithmiq.fi, arXiv, LinkedIn, Crunchbase, 行业媒体

一、基本信息

字段	内容
公司名称	Algorithmiq
中文名	算法量子
成立年份	2020
总部	Milan, Italy (原 Helsinki, Finland)
网站	https://algorithmiq.fi
员工数	50
CEO	Sabrina Maniscalco
量子比特类型	软件层—硬件无关 (支持超导/离子阱/中性原子)
技术阶段	早期商业化 (2024 年起)
商业模式	量子算法软件 + 企业合作 (B2B SaaS + 联合研发)

二、融资历史

融资总额: $42M

日期	轮次	金额 (USD)	投资方	备注
2021	种子轮	$4M	未公开	初始启动资金
2023	A 轮	$15M	Inventure VC	约 €13.7M
2025	非稀释性奖金	$2M	Wellcome Leap	Q4Bio Challenge 唯一冠军 ($50M 总奖金池)
2026-05	B 轮	$21M	United Ventures, CDP Venture Capital, Inventure VC	€18M, 意大利最大量子 VC 轮。总部迁至米兰

三、团队详情（41 人）

1. Sabrina Maniscalco

职位: CEO and Co-founder LinkedIn: https://www.linkedin.com/in/sabrina-maniscalco-0402642/ Google Scholar: https://scholar.google.com/citations?user=EZOGPwgAAAAJ

学历: PhD in Physics —Quantum Information Physics (2004) 毕业院校: University of Palermo, University of Helsinki, Aalto University

职业经历: - University of Helsinki: Professor of Quantum Information, Computing and Logic (2020-present) - University of Turku: Chair of Theoretical Physics (2014-2020) - Heriot-Watt University: Associate Professor (Prior to 2014) - University of Sofia: Postdoctoral Researcher (2004-2006) - University of Durban: Postdoctoral Researcher (2006-2008)

研究方向: 开放量子系统与非马尔可夫动力学, 量子退相干与纠缠, 近期量子设备上的量子模拟, 量子错误缓解, 量子信息理论

在 Algorithmiq 的角色: CEO 兼联合创始人。领导公司战略、融资和合作伙伴关系（IBM、Cleveland Clinic、CERN、NVIDIA）。推动将量子算法应用于生命科学和药物发现的使命。h-index: 53 职业阶段: 资深学术领袖兼初创公司 CEO。赫尔辛基大学正教授，拥有 20 多年量子技术经验。

代表论文: - Sudden transition between classical and quantum decoherence (2010, 641 citations) - Protecting entanglement via the quantum Zeno effect (2008, 550 citations) - Non-Markovian quantum jumps (2008, 395 citations) - Degree of non-Markovianity of quantum evolution (2014, 370 citations)

主要成就: - 世界经济论坛技术先锋（2024 年，与 Algorithmiq 共同获得）- Wellcome Leap Q4Bio 200 万美元奖金得主（2026 年）——唯一提供用于药物发现的可扩展端到端量子框架的团队 - 芬兰量子技术卓越中心副主任 - 发表 250 多篇论文 - 为 Algorithmiq 筹集了超过 3600 万美元资金

2. Guillermo García-Pérez

职位: CSO and Co-founder LinkedIn: https://www.linkedin.com/in/guillermo-garc%C3%ADa-p%C3%A9rez-157143b9/

学历: PhD in Physics —Complex Systems and Network Science (2018) 毕业院校: University of Barcelona, University of Turku

职业经历: - University of Turku: Senior Researcher, Turku Centre for Quantum Physics (2018-2020) - University of Helsinki: Academy of Finland Grant Researcher (2021) - University of Barcelona: PhD Researcher, Dept. de Física Fonamental (2014-2018)

研究方向: 复杂网络与几何重整化, 量子信息科学, 用于错误缓解的张量网络方法, 信息完备测量（IC-POVMs）, 用于生命科学与药物发现的量子算法

在 Algorithmiq 的角色: 首席科学官兼联合创始人。领导科学战略，开发量子算法（TEM、Aurora 平台），监督管理与 IBM Quantum 及其他硬件提供商的研究合作。h-index: 18-25 职业阶段: 职业生涯早期至中期研究员，正向科学领导角色转型。2018 年获得博士学位，迅速晋升为首席科学官。

代表论文: - Multiscale unfolding of real networks by geometric renormalization (Nature Physics, 2018) - Learning to Measure: Adaptive Informationally Complete Generalized Measurements for Quantum Algorithms (PRX Quantum, 2021, 87 citations) - The geometric nature of weights in real complex networks (Nature Communications) - Regulation of burstiness by network-driven activation (Nature Scientific Reports, 2015)

主要成就: - 共同开发了 Algorithmiq 的 TEM（张量网络错误缓解）算法 - Algorithmiq Aurora 平台的核心架构师 - 在《Nature Physics》和《Nature Communications》上发表了文章 - 芬兰科学院资助获得者 - IQT NYC 会议演讲人

3. Matteo Rossi

职位: CTO and Co-founder LinkedIn: https://www.linkedin.com/in/matteo-rossi-9ab6b3183/

学历: PhD in Physics —Quantum Information and Quantum Simulation (2017) 毕业院校: University of Milan, University of Parma, University of Helsinki

职业经历: - University of Helsinki: Docent (Adjunct Professor) in Theoretical Physics (2020-present) - University of Turku: Researcher (2017-2020) - University of Nottingham: Researcher (2015-2017)

研究方向: 量子信息理论与应用, 近期设备上的量子模拟, 错误缓解技术, 开放量子系统与量子计量学, 复杂网络上的量子行走

在 Algorithmiq 的角色: 首席技术官兼联合创始人。领导 Algorithmiq 量子软件平台的技术开发，监督工程团队和产品架构。h-index: 25 职业阶段: 职业生涯中期的量子物理学家，拥有出色的发表记录。赫尔辛基大学 Docent（客座教授），同时兼任初创公司 CTO。

代表论文: - Learning to Measure: Adaptive Informationally Complete Generalized Measurements for Quantum Algorithms (PRX Quantum, 2021) - IBM Q Experience as a versatile experimental testbed for simulating open quantum systems - Emergent entanglement structures and self-similarity in quantum spin chains (Phil. Trans. R. Soc. A, 2022) - Quantum hypergraph states

主要成就: - 共同开发了 Algorithmiq 关于信息完备测量的核心知识产权 - 担任 Algorithmiq 技术平台的软件负责人 - 在 111 篇出版物中被引用超过 2,630 次 - 赫尔辛基大学 Docent - QCon 量子大会主旨演讲者

4. Boris Sokolov

职位: Lead Researcher and Co-founder

学历: PhD in Theoretical and Mathematical Physics —Open Quantum Systems and Quantum Networks (2020) 毕业院校: University of Turku, University of Helsinki

职业经历: - University of Helsinki: Researcher, QTF Centre of Excellence (2020-present) - University of Turku: PhD Researcher, Complex Systems Research Group (2016-2020)

研究方向: 开放量子系统与相对论量子信息, 复杂量子网络与多体物理, 自适应信息完备测量, 变分量子算法, 量子自旋链中的纠缠结构

在 Algorithmiq 的角色: 首席研究员兼联合创始人。推动面向化学和药物发现的量子算法研究。专注于近期量子实用性以及将量子计算与生命科学应用相连接。h-index: 8-12 职业阶段: 早期职业研究员。于 2020 年获得博士学位，从学术界过渡到初创公司领导层。

代表论文: - Learning to Measure: Adaptive Informationally Complete Generalized Measurements for Quantum Algorithms (PRX Quantum, 2021, 87 citations) - Emergent entanglement structures and self-similarity in quantum spin chains (Phil. Trans. R. Soc. A, 2022) - Exploring connections between open quantum systems, relativity and complex quantum networks (PhD Thesis, University of Turku, 2020)

主要成就: - Algorithmiq 联合创始人（2020 年）- Algorithmiq 测量优化和错误缓解知识产权的主要贡献者 - 博士论文探讨了开放量子系统、相对论与复杂网络之间的联系 - Wellcome Leap Q4Bio 奖得主（2026 年，作为 Algorithmiq 团队一员）

5. Stefan Knecht

职位: Head of Quantum, R&D Integration and Execution

学历: PhD in Theoretical Chemistry / Habilitation (Venia Legendi) in Theoretical Chemistry —Relativistic Quantum Chemistry and DMRG (2009) 毕业院校: Heinrich Heine University Düsseldorf, ETH Zürich, University of Strasbourg, University of Southern Denmark, Johannes Gutenberg University Mainz

职业经历: - ETH Zürich: Postdoctoral Researcher / Habilitation, Laboratory of Physical Chemistry (Markus Reiher group) (2010-2020) - University of Strasbourg: Postdoctoral Researcher (2009-2010) - University of Southern Denmark, Odense: Postdoctoral Researcher (2008-2009) - Johannes Gutenberg University Mainz: Privatdozent (2020-present)

研究方向: 用于量子化学的密度矩阵重整化群（DMRG）, 相对论量子化学与重元素, 变分量子本征求解器（VQE）和轨道优化方法, 多组态电子结构理论, 用于药物发现的量子嵌入方案

在 Algorithmiq 的角色: 量子化学团队负责人/生命科学中量子化学问题高效量子算法开发部门主管。领导药物发现中量子化学应用的算法开发。h-index: 25-30 (in quantum chemistry; note: some aggregate sites conflate with a namesake neurologist) 职业阶段: 资深量子化学家，已获得特许任教资格。在转向行业领导角色的同时，曾长期担任学术职位（Privatdozent）。

代表论文: - New Approaches for ab initio Calculations of Molecules with Strong Electron Correlation (Chimia, 2016) - Communication: Four-component density matrix renormalization group (J. Chem. Phys., 2014) - Quantum correlations in molecules: from quantum resourcing to chemical bonding (Quantum Sci. Technol., 2023) - Exact two-component Hamiltonians for relativistic quantum chemistry (J. Chem. Phys., 2022)

主要成就: - 从 ETH Zürich 获得理论化学 Venia legendi（特许任教资格）- 首创了四分量相对论 DMRG 方法 - 共同开发了 QCMaquis DMRG 软件 - 超过 290 篇出版物（在与同名者区分前）- 在连接经典量子化学与量子计算方面拥有丰富经验

6. Daniel Cavalcanti

职位: Lead Quantum Information Researcher

学历: PhD in Theoretical Physics —Quantum Information and Quantum Correlations (2008) 毕业院校: ICFO - Institute of Photonic Sciences, Barcelona, Centre for Quantum Technologies, Singapore

职业经历: - ICFO - Institute of Photonic Sciences: Ramón y Cajal Fellow / Senior Researcher (2013-2021) - Centre for Quantum Technologies, Singapore: Postdoctoral Researcher / Independent Researcher (2010-2013) - ICFO - Institute of Photonic Sciences: Postdoctoral Researcher (2009)

研究方向: 贝尔非定域性与网络中的量子关联, 量子导引与纠缠认证, 量子隐形传态与测量不相容性, 量子信息的半定规划, 用于量子态估计的张量网络方法

在 Algorithmiq 的角色: 首席量子信息研究员。运用量子信息理论开发药物发现算法。还设计了公司的视觉形象和品牌。h-index: 35-40 职业阶段: 资深高级研究员，拥有超过 15 年的博士后经验。从终身教职转向产业界。

代表论文: - All Entangled States can Demonstrate Nonclassical Teleportation (Physical Review Letters, 2017) - Detection of entanglement in asymmetric quantum networks and multipartite quantum steering (Nature Communications, 2015) - Quantitative relations between measurement incompatibility, quantum steering, and nonlocality (Physical Review A, 2016) - Bell nonlocality (Reviews of Modern Physics, 2014 - co-author)

主要成就: - 合著教科书《Semidefinite Programming in Quantum Information Science》(IOP, 2023) - Ramón y Cajal 著名研究基金（西班牙）- 108 篇出版物，约 6000 次引用 - 使用真实量子计算机的输出设计了 Algorithmiq 的企业品牌形象 - 还拥有平面设计硕士学位，并运营 Bitflow 设计工作室

7. Fabijan Pavosevic

职位: Lead Quantum Chemist Google Scholar: https://scholar.google.com/citations?user=q2lveLEAAAAJ

学历: PhD in Theoretical Chemistry —Theoretical and Quantum Chemistry (2017) 毕业院校: Virginia Tech, Yale University, Flatiron Institute (Center for Computational Quantum Physics), University of Zagreb

职业经历: - Flatiron Institute: Flatiron Research Fellow, Center for Computational Quantum Physics (CCQ) (2021-2023) - Yale University: Postdoctoral Associate (Sharon Hammes-Schiffer group) (2017-2021) - Virginia Tech: PhD Researcher (Edward Valeev group) (2012-2017) - University of Zagreb: BS/MS Student, Faculty of Chemical Engineering and Technology (2003-2009)

研究方向: 核-电子轨道（NEO）方法, 量子计算的耦合簇理论, 幺正耦合簇（UCC）与 ADAPT-VQE, 极化激元化学与腔修饰反应, 活性空间选择（AEGISS 工作流）

在 Algorithmiq 的角色: 首席/主力量子化学家。为近期量子设备开发量子化学算法，重点研究用于药物发现和材料科学的分子模拟。h-index: 16 职业阶段: 早期至中期职业研究员。2017 年完成博士学位，在加入 Algorithmiq 之前担任博士后和研究助理职位。

代表论文: - Software for the frontiers of quantum chemistry: Q-Chem 5 (2021, 841 citations) - Multicomponent quantum chemistry via the NEO method (Chemical Reviews, 2020, 126 citations) - Error-Mitigation Enabled Multicomponent Quantum Simulations Beyond the Born-Oppenheimer Approximation (J. Chem. Theory Comput., 2026) - Spin-Flip Unitary Coupled Cluster Method: Toward Accurate Description of Strong Electron Correlation on Quantum Computers (J. Phys. Chem. Lett., 2023)

主要成就: - 计算量子物理学中心 Flatiron 研究所研究员 - Q-Chem 5 软件的关键贡献者（841 次引用）- 开发了用于量子计算机上的量子化学的 AEGISS 活性空间选择工作流 - 在量子计算机上进行后玻恩-奥本海默分子模拟的开创性工作

8. Martina Stella

职位: Senior Researcher, Head of Partnerships Google Scholar: Not publicly indexed; profile exists per AD Scientific Index

学历: PhD in Theoretical and Computational Chemistry —Theoretical and Computational Chemistry (2015) 毕业院校: University of Bristol, University of Cambridge, University of Naples

职业经历: - Imperial College London: Researcher (2018-2022) - King’s College London: Postdoctoral Researcher (2015-2018) - ICTP (Abdus Salam International Centre for Theoretical Physics): Boltzmann Senior Fellow / Long-term Visiting Scientist (2022-present)

研究方向: 用于化学和生命科学的量子计算，肿瘤光动力疗法（PDT），激发态的密度泛函理论（DFT）方法，BigDFT 代码开发，活性空间选择（AEGISS 方法）

在 Algorithmiq 的角色: 高级研究员，合作伙伴关系负责人——识别量子计算机可以模拟的生物系统，为生命科学开发 NISQ 软件 h-index: 10 (total), 9 (recent, last 5 years) 职业阶段: 早期至中期职业研究人员

代表论文: - AEGISS —Atomic orbital and Entropy-based Guided Inference for Space Selection (arXiv, 2025) - Scalable quantum circuit generation using Majorana Propagation (ADAPT-VMPE) - Data-Efficient Active Learning Discovery of Transition Metal Photosensitizers for Type I PDT - Aqueous Solution Chemistry In Silico and the Role of Data Driven Approaches

主要成就: - 双学术-产业职位：在 Algorithmiq 全职工作的同时保持 ICTP 的从属关系 - 在 ICTP 获得 Boltzmann Fellowship 独立学术职位 - BigDFT 代码的共同开发者 - 总引用约 938 次，近期引用约 640 次

9. Sergei Filippov

职位: Senior Researcher Google Scholar: https://scholar.google.com/citations?user=AimFBr8AAAAJ&hl=en

学历: PhD in Theoretical Physics (2012) —Theoretical Physics —Quantum Information Theory (2012) 毕业院校: Moscow Institute of Physics and Technology (MIPT)

职业经历: - Moscow Institute of Physics and Technology (MIPT): Associate Professor (-) - Steklov Mathematical Institute, Russian Academy of Sciences: Senior Researcher (2018-)

研究方向: Open quantum systems and non-Markovian dynamics, Quantum error mitigation via tensor networks, Quantum measurements and tomography, Quantum entanglement and quantum channels, Quantum communication (coherent information superadditivity), Machine learning for quantum dynamics

在 Algorithmiq 的角色: Senior Researcher —leads tensor network noise characterization and error mitigation research h-index: 15-20 (estimated; ~70 papers, ~1,000 citations on Rankless) 职业阶段: Established researcher

代表论文: - Scalable tensor-network error mitigation for near-term quantum computing (arXiv:2307.11740) —foundational TEM method - Machine learning non-Markovian quantum dynamics (Physical Review Letters, 2020) —79 citations - Divisibility of quantum dynamical maps and collision models (Physical Review A, 2017) —70 citations - Simulation Complexity of Open Quantum Dynamics: Connection with Tensor Networks (Physical Review Letters, 2019) —57 citations

主要成就: - Core inventor of Algorithmiq’s Tensor-network Error Mitigation (TEM) method, commercially launched on IBM Qiskit Functions Catalog (Sept 2024) - Co-author with Alexander S. Holevo on quantum Gaussian maximizers - Born June 19, 1987

10. John Goold

职位: Lead Algorithmiq Ireland Google Scholar: https://scholar.google.com/citations?user=uPtQT_sAAAAJ

学历: PhD in Physics (2010) —Physics —Quantum Thermodynamics (2010) 毕业院校: University College Cork

职业经历: - Trinity College Dublin: Full Professor of Physics (2024-present) - Trinity College Dublin: Associate Professor (2019-2024) - Trinity College Dublin: Assistant Professor (2017-2019) - ICTP (Abdus Salam International Centre for Theoretical Physics): Research Scientist (2013-2017) - University of Oxford, Clarendon Laboratory: INSPIRE Marie Curie Fellow (2010-2013)

研究方向: 量子热力学，量子信息理论，多体物理，量子电池与充电，非平衡量子引擎

在 Algorithmiq 的角色: Algorithmiq 爱尔兰负责人——连接 Algorithmiq 与都柏林圣三一学院生态系统，合作使用 IBM 硬件进行错误缓解实验 h-index: 37 职业阶段: 资深/成熟研究员（正教授）

代表论文： - The role of quantum information in thermodynamics—a topical review (2016) —1,048 citations - Experimental reconstruction of work distribution and study of fluctuation relations in a closed quantum system (2014) —432 citations - Enhancing the charging power of quantum batteries (2017) —376 citations - Quantacell: powerful charging of quantum batteries (2015) —373 citations

主要成就： - 与合作伙伴 IBM、微软、Algorithmiq、Horizon Quantum、Moody's Analytics 共同创立三一量子联盟（TQA）- 在都柏林圣三一学院创办爱尔兰首个量子科学与技术硕士项目 - 当选都柏林圣三一学院院士（2022）- 使用 Algorithmiq TEM 在 IBM 127 量子比特 Eagle 处理器上进行实验，并在 IBM 量子峰会上展示 - SFI-皇家学会大学研究员

11. Caterina Foti

职位： Senior Researcher and Outreach Coordinator LinkedIn: https://www.linkedin.com/in/caterina-foti/ Google Scholar: https://orcid.org/0000-0002-3977-4295

学历： PhD in Physics —Quantum Physics / Foundations of Quantum Mechanics (2019) 毕业院校： University of Florence (PhD, 2019), Aalto University (Postdoctoral Researcher)

职业经历： - Aalto University: Postdoctoral Researcher, Department of Applied Physics (2020-present) - Aalto University: Intern/Visiting Researcher (during PhD) (2017) - Algorithmiq Ltd: Senior Researcher and Outreach Coordinator (2020-present) - University of Pisa: Research Collaborator ()

研究方向： 量子物理教育与外展，面向科学素养的量子游戏，量子物理学基础（Page and Wootters mechanism），从量子纠缠中产生时间的机制，文化-科学故事叙述

在 Algorithmiq 的角色： 高级研究员与外展协调员-负责量子外展、QPlayLearn 开发、品牌推广和基础量子研究 h-index: ~8-10 (based on Nature Communications paper with ~12 Scopus citations plus education papers) 职业阶段： 职业生涯中期（2019 年获博士学位，博士后与产业双重角色）

代表论文： - Time and classical equations of motion from quantum entanglement via the Page and Wootters mechanism with generalized coherent states (Nature Communications, 2021) - Games for Quantum Physics Education (Journal of Physics: Conference Series, 2024) - Quantum games - a way to shed light on quantum mechanics (Europhysics News, 2022) - Quantum Physics Literacy Aimed at K12 and the General Public (Universe, 2021)

主要成就： - 基于游戏和互动工具的在线量子素养平台 QPlayLearn 的共同创建者 - 量子丛林互动展览（在比萨 Palazzo Blu 展出）的共同创建者 - 为公众参与组织了量子游戏创作营 - 被阿尔托大学新闻专题报道：'我的梦想是让 0 到 99 岁的人都能接触量子世界'- 芬兰科学院资助项目（FQF P5 T30460，2024-2028 年）

12. Roberto Di Remigio Eikås

职位： Head of Product Engineering LinkedIn: https://www.linkedin.com/in/robertodr/ Google Scholar: https://scholar.google.com.tr/citations?user=GO15fgsAAAAJ

学历： PhD —Theoretical and Computational Chemistry (Quantum Molecular Sciences) (2017) 毕业院校： UiT The Arctic University of Norway (Tromso), University of Pisa

职业经历： - UiT The Arctic University of Norway / Hylleraas Centre for Quantum Molecular Sciences: Postdoctoral Fellow (2017-2020) - Virginia Tech: Visiting Researcher (Prior to 2020) - University of Pisa - Molecular Modelling Lab: Researcher / Contributor (Prior to 2022) - Algorithmiq Ltd: Lead Software Engineer, R&D (2022-2024)

研究方向： 量子化学软件开发（PCMSolver、DIRAC、PSI4、MRChem、VAMPyR），连续溶剂化模型（极化连续介质模型），相对论量子化学（狄拉克-库仑-布莱特哈密顿量），耦合簇理论与图解蒙特卡罗方法，用于电子结构的多小波方法

在 Algorithmiq 的角色： 产品工程负责人（曾任首席软件工程师，研发岗）——连接量子化学研究与产品开发，主导 Algorithmiq 量子计算平台的软件工程 h-index: 12-13 (24+ publications, ~2000+ total citations) 职业阶段： 中高级职业阶段（博士 + 博士后 +4 年以上行业经验，量子化学软件领域的公认领导者）

代表论文： - Full Breit Hamiltonian in the Multiwavelets Framework - J. Chem. Theory Comput. (2024) - VAMPyR - A high-level Python library for mathematical operations in a multiwavelet representation - J. Chem. Phys. (2024) - Cavity-Free Continuum Solvation: Implementation and Parametrization in a Multiwavelet Framework - J. Chem. Theory Comput. (2023) - MRChem Multiresolution Analysis Code: Performance and Scaling Properties - J. Chem. Theory Comput. (2022)

主要成就: - PCMSolver（溶剂化建模开源库）的主要开发者 - 多个旗舰量子化学程序（PSI4、DIRAC、Dalton Project）的重要贡献者 - 科学计算领域广泛使用的《CMake Cookbook》合著者 - 2023 年北欧量子秋季学校特邀讲者（变分量子本征求解器与量子化学）- 积极参与开源科学软件的可持续性与互操作性工作

13. Kirsten Nehr

职位: Chief Operating Officer LinkedIn: https://www.linkedin.com/in/kirstennehr/ Google Scholar: N/A

学历: Master’s degree —Biological Sciences (Not publicly specified) 毕业院校: University of Oxford

职业经历: - Healthcare and Technology sectors: Commercial roles (marketing, branding, life sciences partnerships, business strategy, operations) (10+ years)

在 Algorithmiq 的角色: 首席运营官（COO）/ 幕僚长——在专注于医疗/生命科学的量子计算与人工智能公司中，负责运营、业务战略、市场营销、品牌建设及生命科学合作伙伴关系 h-index: N/A 职业阶段: 高级/高管（在医疗和科技领域拥有超过 10 年的商业经验）

主要成就: - 在 AI for Good（国际电信联盟）平台发表演讲 - 在 Algorithmiq 打造定义行业类别的量子计算业务

14. Ben Bakker

职位: Chief Financial Officer LinkedIn: https://www.linkedin.com/in/benjaminbakkernz/ Google Scholar: N/A

学历: Not publicly available —Not publicly available (likely Finance / Business) (Not publicly available)

在 Algorithmiq 的角色: 首席财务官 - 负责 Algorithmiq 的财务战略、融资及财务运营 h-index: N/A 职业阶段: 高级 / 高管 (CFO 级别)

主要成就: - 量子计算初创公司 Algorithmiq 的首席财务官

15. Irene Gentili

职位: Chief of Staff LinkedIn: https://www.linkedin.com/in/irenegentili/ Google Scholar: N/A

学历: Master’s degree —Business / Management (Not publicly specified) 毕业院校: Bocconi University (Milan) - Bachelor’s, ESADE Business School (Barcelona) - Master’s

职业经历: - Redstone: VC Investor (~2.5 years) - Speedinvest: Associate, Marketplaces & Consumer team () - Samaipata: VC Investor () - Consulting (Milan): Consultant (Early career)

在 Algorithmiq 的角色: 幕僚长 - 负责构建运营架构、流程和日常运营，以支持公司规模化发展阶段。常驻米兰（Algorithmiq 全球总部）。h-index: N/A 职业阶段: 从风险投资转型至初创企业运营领导层的早中期职业专业人士

主要成就: - 登上《Woman Taught Tech》播客，探讨风险投资领域的多样性和对女性创始人的资金支持 - 从风险投资领域转型至深科技量子初创企业的运营领导层

16. Minna Gunes

职位: Head of People and Strategic Funding LinkedIn: https://www.linkedin.com/in/minna-gunes/ Google Scholar: N/A (limited publication record)

学历: PhD in Natural Sciences —Natural Sciences (2004) 毕业院校: University of Jyvaskyla - PhD (2004)

职业经历: - Aalto University: Academic Coordinator for InstituteQ, Quantum Technology Finland, OtaNano, Centre for Quantum Engineering (~2015-2024) - University of Helsinki: Researcher/Coordinator ()

研究方向: 量子生态系统协调与基础设施管理

在 Algorithmiq 的角色: 人才与战略资助负责人（在 Algorithmiq 团队页面上也同时列为战略联盟负责人兼人才合伙人）——负责人才运营、战略伙伴关系和资助协调 h-index: N/A (career focused on academic coordination rather than primary research) 职业阶段: 从学术协调和量子生态管理转向产业人力资源与战略合作伙伴关系的高级专业人士

代表论文: - InstituteQ - The Finnish National Quantum Institute - Arkhimedes (2022), co-authored with J.P. Pekola, H.S. Majumdar, P. Muratore Ginanneschi

主要成就: - 芬兰国家量子生态系统计划（InstituteQ、Quantum Technology Finland、OtaNano）的关键协调人 - 多语能力：芬兰语（母语），英语（流利），瑞典语（一般），德语/西班牙语（基础），土耳其语（良好口语，书面语一般）- 代表 Algorithmiq 参与北欧-波罗的海量子生态系统报告（Nordic Innovation）

17. Emilia Silius

职位: Business Operations Coordinator LinkedIn: https://www.linkedin.com/in/emilia-silius/ Google Scholar: N/A

学历: Master’s degree —Marketing / Business (2024) 毕业院校: Aalto University School of Business - Master’s (2024), Aalto University School of Business - Bachelor’s in Marketing (2021)

职业经历: - Aalto University: Student (Bachelor’s and Master’s theses) (2021-2024)

研究方向: Consumer behavior and retail therapy (Bachelor’s thesis), Slow fashion and sustainable consumption (Master’s thesis)

在 Algorithmiq 的角色: Business Operations Coordinator - responsible for business operations support at the Helsinki office h-index: N/A 职业阶段: Early career professional, likely first role after completing Master’s degree in 2024

代表论文: - Bachelor’s thesis: ‘Can Money Buy Happiness?’ - A literature review on retail therapy (2021) - Master’s thesis: ‘Fashion is temporary, but style endures: Exploring the slow fashion phenomenon and efforts of sustainable clothing brands toward more sustainable consumption’(2024)

主要成就: - Completed both Bachelor’s and Master’s degrees at Aalto University School of Business - Master’s thesis involved qualitative research with Nordic sustainable fashion brands

18. Tommaso Tufarelli

职位: Senior Researcher and R&D Project Manager Google Scholar: Not found in public search

学历: PhD (Theoretical Quantum Physics) —Quantum Optics and Open Quantum Systems () 毕业院校: University of Nottingham, Imperial College London, University College London, Universitat Ulm

职业经历: - University of Nottingham: Assistant Professor, School of Mathematical Sciences (~2021-present) - Imperial College London: Researcher, QOLS Blackett Laboratory (-) - University College London: Researcher, Department of Physics & Astronomy (-) - Universitat Ulm: Researcher, Institut fur Theoretische Physik (-)

研究方向: 量子光学与开放量子系统, 非马尔可夫量子动力学, 量子计量学与量子费希尔信息, 量子关联与量子失谐, 腔量子电动力学与电路量子电动力学

在 Algorithmiq 的角色: 高级研究员兼研发项目经理（用户所述职务；公开网络无直接确认其在 Algorithmiq 任职的信息——公开资料显示其为诺丁汉大学助理教授）h-index: Not available —likely moderate based on publication record 职业阶段: 职业生涯中期（助理教授 / 高级研究员）

代表论文: - General Expressions for the Quantum Fisher Information Matrix with Applications to Discrete Quantum Imaging (PRX Quantum, 2021) - Coherently opening a high-Q cavity (Physical Review Letters, 2013) - Non-Markovianity of a quantum emitter in front of a mirror (Physical Review A, 2014) - There is more to quantum interferometry than entanglement (Physical Review A, 2017)

主要成就: - 因量子关联与不确定性方面的工作登上 Physical Review Letters（2013 年）封面 - 量子计量学和费希尔信息理论专家

19. Adam Glos

职位: Computer Science Researcher Google Scholar: https://scholar.google.com/citations?user=3f3cR0AAAAAJ

学历: PhD in Computer Science —Computer Science —Quantum Walks and Quantum Algorithms () 毕业院校: Institute of Theoretical and Applied Informatics, Polish Academy of Sciences, Silesian University of Technology

职业经历: - Institute of Theoretical and Applied Informatics, Polish Academy of Sciences: PhD Researcher (-) - QWorld Association: QResearch Coordinator (-)

研究方向: 图上的量子行走与空间搜索, 量子优化（QAOA，量子退火）, 变分量子算法, 量子电路编译与态制备, 量子网络医学

在 Algorithmiq 的角色: 计算机科学研究员—态制备团队负责人，专注于 NISQ 设备的电路编译与优化 h-index: 12-15 (estimated from Google Scholar citation counts) 职业阶段: 早期至中期职业研究员

代表论文: - Prog-QAOA: Framework for resource-efficient quantum optimization through classical programs (Quantum 9:1663, 2025) - Treespilation: Architecture- and State-Optimised Fermion-to-Qubit Mappings - A generic multi-Pauli compilation framework for limited connectivity (2024) - Quantum Optimization for the Graph Coloring Problem with Space-Efficient Embedding (QCE, 2020) —53 citations

主要成就: - 与 Ozlem Salehi Koken 在 Classiq 编码挑战赛（2022）中获得金牌 - Algorithmiq 公司态制备团队负责人 - 在 arXiv 上发表了 50 多篇论文 - 共同开发了用于量子随机行走分析的 QSWalk.jl 软件包

20. Walter Talarico

职位: Senior Researcher Google Scholar: https://scholar.google.co.th/citations?user=I9J5N84AAAAJ

学历: PhD in Theoretical Physics (2020) —Theoretical Physics —Condensed Matter and Quantum Many-Body Physics (2020) 毕业院校: University of Turku, Universita degli Studi della Calabria

职业经历: - Aalto University: Postdoctoral Researcher, Department of Applied Physics (2021-2023) - University of Turku: PhD Student, Department of Physics and Astronomy (2017-2020)

研究方向: 化学领域的量子算法（VQE、ADAPT-VQE、qEOM）, 量子多体物理与开放量子系统, 非平衡格林函数, 纳米尺度系统中的量子输运, 变分量子本征求解器方法

在 Algorithmiq 的角色: 高级研究员—为近期量子计算机设计优化软件和算法，专注于量子化学应用 h-index: 3-6 (estimated from partial citation data) 职业阶段: 职业生涯早期研究员（2020 年获博士学位，2021 年加入 Algorithmiq）

代表论文: - Toward Accurate Post-Born-Oppenheimer Molecular Simulations on Quantum Computers (J. Chem. Theory Comput., 2023) - Self-Consistent Field Approach for the Variational Quantum Eigensolver: Orbital Optimization Goes Adaptive (J. Phys. Chem. A, 2024) - The variational quantum eigensolver self-consistent field method within a polarizable embedded framework (J. Chem. Phys., 2024) - Estimating molecular thermal averages with the quantum equation of motion and informationally complete measurements (Entropy, 2024)

主要成就: - 开发了具有可极化嵌入的 VQE-SCF 方法，用于真实化学体系 - 在量子中心的强关联和动态电子关联方面做出贡献（面向近期设备的 NEVPT2）- 开放量子系统和非平衡格林函数专家

21. Zoltán Zimborás

职位: Senior Researcher Google Scholar: Not directly found via public search (likely exists but URL not indexed)

学历: PhD —Quantum Information Science () 毕业院校: Eotvos University, Budapest, UPV Bilbao, University College London, Freie Universitat Berlin

职业经历: - Wigner Research Centre for Physics, Budapest: Leader, Quantum Computing and Information Research Group (-) - Eotvos Lorand University: Faculty of Informatics (-) - University of Helsinki: Researcher, Department of Physics (-) - UPV Bilbao: Postdoctoral Researcher (-) - University College London: Postdoctoral Researcher (-)

研究方向：量子门分解与编译，量子优势方案与计算普适性，变分量子算法，量子误差缓解（读出噪声），分子中的量子关联

在 Algorithmiq 的角色：高级研究员——量子算法设计、误差缓解与门分解 h-index: 15-20 (estimated from publication citation counts) 职业阶段：中高级职业研究人员

代表论文：- Approaching the theoretical limit in quantum gate decomposition (Quantum 6:710, 2022) - Mitigation of readout noise in near-term quantum devices by classical post-processing based on detector tomography (Quantum 4:257, 2020) - Fermion Sampling: A Robust Quantum Computational Advantage Scheme (PRX Quantum, 2022) - Error mitigation for variational quantum algorithms through mid-circuit measurements (Phys. Rev. A, 2022) —45 citations

主要成就：- QWorld（非营利性量子编程教育组织）董事会成员 - NKFI 资助项目“近期量子计算机”首席研究员（2020-2025，约 3550 万匈牙利福林）- 匈牙利量子信息国家实验室（QNL）成员 - 接近理论下界的最优量子门分解专家

22. Pi Haase

职位：Computational Chemistry Researcher LinkedIn: https://www.linkedin.com/in/pi-a-b-haase/ Google Scholar: https://arxivlens.com/Author/Details/pi-a.-b.-haase-9162-5883937

学历：PhD in Chemistry —Relativistic Quantum Chemistry / Electronic Structure Theory (2021) 毕业院校：University of Groningen (PhD, 2021), University of Copenhagen (BSc/MSc, likely)

职业经历： - Cleveland State University: Visiting/Research Affiliate (2024-2025) - University of Groningen: PhD Researcher (2017-2021) - UiT The Arctic University of Norway: Research Collaborator (prior to 2021)

研究方向：相对论耦合簇方法，超精细结构常数与不确定度估计，分子宇称不守恒，多参考量子化学的活性空间选择（AEGISS 方法），量子计算在化学中的应用

在 Algorithmiq 的角色：计算化学研究员——开发用于分子模拟中量子计算应用的量子化学方法 h-index: Not publicly available (early career, ~15 publications) 职业阶段：早期至中期职业研究者（生于 1990 年，2021 年获博士学位）

代表论文：- AEGISS -Atomic orbital and Entropy-based Guided Inference for Space Selection (arXiv, 2025) - Hyperfine Structure Constants on the Relativistic Coupled Cluster Level with Associated Uncertainties (J. Phys. Chem. A, 2020) - Benchmarking of the Fock space coupled cluster method and uncertainty estimation: Magnetic hyperfine interaction in the excited state of BaF - Towards detection of the molecular parity violation in chiral Ru(acac)3 and Os(acac)3

主要成就：- 开发了 AEGISS 方法，用于半自动活性空间选择，将轨道熵分析与原子轨道投影统一起来 - 提出分子性质计算的第一性原理不确定度估计方案，可实现约 5.5% 的保守不确定度界限 - 架起量子化学与药物发现中量子计算应用之间的桥梁

23. Özlem Salehi

职位：Computer Science Researcher LinkedIn: https://www.linkedin.com/in/ozlem-salehi/ Google Scholar: https://ieeexplore.ieee.org/author/37088516883

学历：PhD in Theoretical Computer Science —Computer Engineering / Theoretical Computer Science (2019) 毕业院校：Bo aziçi University, Istanbul (PhD, 2019), Bo aziçi University, Istanbul (MSc, 2013)

职业经历：- Institute of Theoretical and Applied Informatics, Polish Academy of Sciences (IITiS PAN): Postdoctoral Researcher (2021-2023) - Özye in University: Instructor, Department of Computer Science (2019-2021) - Algorithmiq: Computer Science Researcher (2023-present)

研究方向：量子优化（QAOA、量子退火、VQE），用于组合优化的 QUBO 和 HOBO 公式化，量子计算教育，量子有限自动机与计算复杂性

在 Algorithmiq 的角色：计算机科学研究员——开发用于生命科学应用的量子优化算法 h-index: ~5 (384 total citations per AD Scientific Index) 职业阶段：职业生涯中期（2019 年博士毕业，有博士后经历，现于工业界）

代表论文: - Prog-QAOA: Framework for resource-efficient quantum optimization through classical programs (Quantum, Vol. 9, 2025) - Optimizing the Production of Test Vehicles Using Hybrid Constrained Quantum Annealing (SN Computer Science, 2023) - Unconstrained Binary Models of the Travelling Salesman Problem Variants for Quantum Optimization (Quantum Information Processing, 2022) -51 citations - Quadratic and Higher-Order Unconstrained Binary Optimization of Railway Rescheduling for Quantum Computing (Quantum Information Processing, 2022)

主要成就: - QTurkey 联合创始人，该社区旨在提升土耳其的量子技术意识 - 组织量子编程工作坊并开发开源教育材料 - 最高引用量子优化论文被引用 51 次 - 在 Sabanc University 的院长特邀讲座研讨会（2025 年）

24. Alessio Fallani

职位: Researcher LinkedIn: https://www.linkedin.com/in/alessiofallani/ Google Scholar: https://orbilu.uni.lu/pro

学历: PhD in Physics —Quantum Physics / Machine Learning for Drug Discovery (2025) 毕业院校: University of Luxembourg (PhD, 2025), University of Milan, Dipartimento di Fisica Aldo Pontremoli

职业经历: - Algorithmiq Ltd: Researcher (2024-present) - University of Luxembourg: PhD Researcher (supervisor: Prof. Alexandre Tkatchenko) (2021-2025) - University of Milan: Research Affiliate (prior to 2021)

研究方向: 用于药物发现的量子化学和机器学习，用于分子设计的量子逆映射（QIM），在量子性质上预训练的图变换器用于 ADMET 建模，用于过渡金属光敏剂发现的主动学习，光子量子计算模拟非共价相互作用

在 Algorithmiq 的角色: 研究员——将量子化学和机器学习应用于药物发现 h-index: ~4 (early career, Nature Communications paper) 职业阶段: 职业早期（1995 年 8 月出生，2025 年 3 月完成博士学位）

代表论文: - Inverse Mapping of Quantum Properties to Structures for Chemical Space of Small Organic Molecules (Nature Communications, 2024) - Pretraining Graph Transformers with Atom-in-a-Molecule Quantum Properties for Improved ADMET Modeling (Journal of Cheminformatics, 2025) - Data-Efficient Active Learning Discovery of Transition Metal Photosensitizers for Type I Photodynamic Therapy (arXiv, 2025) - Modeling Noncovalent Interatomic Interactions on a Photonic Quantum Computer (Physical Review Research, 2023)

主要成就: - 关于量子逆映射分子设计的 Nature Communications 论文 - 由 Janssen Pharma 联合指导的博士研究（产业合作）- 开发了用于类药分子量子力学探索的 Aquamarine（AQM）数据集 - 架起量子计算、机器学习与药物研究的桥梁

25. Fabian Langkabel

职位: Researcher LinkedIn: https://www.linkedin.com/in/fabian-langkabel-728bba275/ Google Scholar: Not publicly found; indexed via Semantic Scholar and arXiv

学历: Dr. rer. nat. (PhD) in Chemistry —Theoretical Chemistry / Quantum Computing (2023) 毕业院校: Freie Universität Berlin (PhD, 2023), Helmholtz-Zentrum Berlin (PhD research), Humboldt-Universität zu Berlin (Master’s)

职业经历: - Algorithmiq Ltd: Researcher (2024-present) - Helmholtz-Zentrum Berlin (HZB): PhD Researcher (HEIBRiDs fellowship) (2020-2023) - Freie Universität Berlin: Doctoral Candidate, Institute of Chemistry and Biochemistry (2020-2023)

研究方向: 分子中电子动力学的量子算法，多分辨率变分量子本征值求解器（VQE），激光驱动的电子动力学模拟，Jellyfish 模块化模拟代码开发，粒子间库仑衰变（ICD）动力学

在 Algorithmiq 的角色: 研究员-将量子算法应用于计算化学，以实现药物研发中的分子模拟 h-index: ~4-5 (early career) 职业阶段: 职业早期（2023 年博士毕业，首份产业界职位）

代表论文: - The advent of fully variational quantum eigensolvers using a hybrid multiresolution approach (arXiv, 2024) - Jellyfish: A modular code for wave function-based electron dynamics simulations and visualizations on traditional and quantum compute architectures (WIREs Computational Molecular Science, 2023) - Quantum-Compute Algorithm for Exact Laser-Driven Electron Dynamics in Molecules (J. Chem. Theory Comput., 2022) - Three-electron dynamics of the interparticle Coulombic decay with two-dimensional continuum confinement (J. Chem. Phys., 2021)

主要成就：- 首次实现分子中精确激光驱动电子动力学的量子计算算法 - 开发了支持经典与量子架构的 Jellyfish 模拟代码 - 入选 HZB 新闻稿（2022 年 11 月）“量子算法节省电子动力学计算时间” - 博士论文：传统与量子计算架构上的从头算电子动力学开发与应用

26. Anton Nykänen

职位：Researcher LinkedIn：https://www.linkedin.com/in/anton-nykänen-6b434a203/ Google Scholar：https://orcid.org/0009-0003-1949-3028

学历：MSc in Theoretical Physics —Theoretical Physics / Quantum Computing (2021) 毕业院校：University of Turku (MSc, 2021), University of Turku (BSc)

职业经历：- Algorithmiq Oy: Junior Researcher / Researcher (2021-present)

研究方向：变分量子本征求解器（VQE）和 ADAPT-VQE 方法，量子计算机上分子的激发能计算，后玻恩-奥本海默分子模拟，费米子到量子位的映射与马约拉纳传播，量子错误缓解策略

在 Algorithmiq 的角色：Algorithmiq 的研究员（初级研究员）——开发用于药物发现中分子模拟的量子算法 h-index: ~4-5 (early career, multiple JPC/JCTC publications) 职业阶段：早期职业阶段（2021 年获得硕士学位，2021 年 6 月加入 Algorithmiq）

代表论文：- Scalable quantum circuit generation for iterative ground state approximation using Majorana Propagation (arXiv, 2026) - Simulation of Fermionic circuits using Majorana Propagation (arXiv, 2025) - Toward Accurate Calculation of Excitation Energies on Quantum Computers with Delta-ADAPT-VQE: A Case Study of BODIPY Derivatives (J. Phys. Chem. Lett., 2024) - Self-Consistent Field Approach for the Variational Quantum Eigensolver: Orbital Optimization Goes Adaptive (J. Phys. Chem. A, 2024)

主要成就：- Algorithmiq 的 ADAPT-VQE 方法论的核心贡献者 - 与 AstraZeneca、IBM Research、Trinity College Dublin 和 Cleveland Clinic 的合作 - 硕士论文：利用近期量子计算机探索纠缠现象（University of Turku，2021 年）

27. Keijo Korhonen

职位：Researcher LinkedIn：https://www.linkedin.com/in/keijokorhonen/ Google Scholar：https://orcid.org/0009-0003-2647-3105

学历：MSc in Theoretical and Computational Methods —Quantum Information / Theoretical Physics (ongoing PhD (started 2023)) 毕业院校：University of Helsinki (PhD ongoing, 2023-present), University of Helsinki (MSc, 2022), University of Helsinki (BSc in Physical Sciences, 2020)

职业经历：- Algorithmiq Ltd: Junior Researcher (2022-present) - University of Helsinki: PhD Student, QTF Centre of Excellence (2023-present)

研究方向：近期量子硬件上的高精度测量，随机测量与经典影子，量子测量层析（QMT），基于局部最优对偶框架的影子估计，分子能量估计（VQE）

在 Algorithmiq 的角色：研究员（初级研究员）——为近期量子计算机开发高精度测量技术 h-index: ~3-4 (early career, npj Quantum Information publication) 职业阶段：早期职业（博士生，2022 年 2 月加入 Algorithmiq）

代表论文：- Practical techniques for high precision measurements on near-term quantum hardware: a Case Study in Molecular Energy Estimation (npj Quantum Information, 2025) - Improving shadow estimation with locally optimal dual frames (Physical Review A, accepted 2026) - Enhanced observable estimation through classical optimization of informationally overcomplete measurement data: Beyond classical shadows (Physical Review A, 2024) - Self-consistent quantum measurement tomography based on semidefinite programming (Physical Review Research, 2023)

主要成就：- 在 npj Quantum Information 发表论文（2025 年）- 在 IBM Eagle r3 量子硬件上对 BODIPY 分子进行分子能量估计 - 硕士学位论文：探索在 VQE 中使用自适应 POVM 测量（赫尔辛基大学，2022 年）

28. Matea Leahy

职位: Researcher (Junior Researcher / Researcher Intern) LinkedIn: https://www.linkedin.com/in/matea-l-075993100/ Google Scholar: Not publicly found

学历: MSc in Quantum Science and Technology (ongoing) —Applied and Computational Mathematics (BSc); Quantum Science and Technology (MSc) (Ongoing) 毕业院校: University College Dublin (UCD), Trinity College Dublin (TCD)

职业经历: - Algorithmiq: Researcher Intern / Junior Researcher (2022-present) - Trinity Quantum Alliance: MSc Thesis Collaborator (2022-present)

研究方向: 量子错误缓解, 张量网络方法, 量子算法

在 Algorithmiq 的角色: 初级研究员/研究实习生——从事量子错误缓解和算法开发工作，与 Algorithmiq 的 Helsinki 团队以及 Dublin 的 Trinity Quantum Alliance 合作 h-index: 1-2 (very early career) 职业阶段: 极早期职业生涯——目前正在完成 MSc 学位，在 Algorithmiq 实习，同时为错误缓解方面的第一作者级别出版物做出贡献

代表论文: - Scalable tensor-network error mitigation for near-term quantum computing (arXiv:2307.11740) -co-author - Dynamical simulations of many-body quantum chaos on a quantum computer (2026, EPFL/TCD collaboration)

主要成就: - 首届 Microsoft Ireland Scholarship 获得者，用于攻读 Trinity College Dublin 的量子科学与技术 MSc 学位（2021/22）- 在 WERQSHOP 2024 上展示了可扩展的张量网络错误缓解工作 - 入选 Trinity College Dublin Impact Report 2021-22 - 研究了超过 4000 个 CNOT 门的 91 量子比特量子电路

29. Joonas Malmi

职位: Researcher (Junior Researcher) LinkedIn: https://www.linkedin.com/in/joonas-malmi-1a5466238/ Google Scholar: Not publicly found (MathSciNet author ID: 1624464)

学历: MSc (PhD student) —Theoretical Physics / Quantum Information (MSc 2020; PhD ongoing) 毕业院校: University of Turku (BSc, MSc), University of Helsinki (PhD, ongoing), Aalto University (research collaboration)

职业经历: - Algorithmiq Ltd: Junior Researcher (2022-present) - University of Helsinki —QTF Centre of Excellence: PhD Researcher (2021-present) - InstituteQ —Finnish Quantum Institute: Researcher (2021-present)

研究方向: 量子信息理论, 测量协议（影子估计、对偶框架）, 复杂网络上的量子行走, 可观测量估计与经典影子

在 Algorithmiq 的角色: 初级研究员——从事量子测量协议、影子估计和量子信息理论的研究；多篇带有 Algorithmiq 单位的论文的通讯作者 h-index: 3-4 职业阶段: 早期职业 / 博士生——于赫尔辛基大学攻读博士学位，已有多篇同行评审的出版物

代表论文: - Enhanced observable estimation through classical optimization of informationally over-complete measurement data —beyond classical shadows (Phys. Rev. A 109, 062412, 2024; arXiv:2401.18049) —first author - Improving shadow estimation with locally-optimal dual frames (arXiv:2511.02555, 2025) —co-author - Spatial search by continuous-time quantum walks on renormalized Internet networks (Phys. Rev. Research 4, 043185, 2022) —first author - Virtual linear map algorithm for classical boost in near-term quantum computing (arXiv:2207.01360, 2022) —co-author

主要成就: - Physical Review A 和 Physical Review Research 论文的第一作者 - 开发了针对对偶 POVM 算子的测量后优化方法，在经典影子导致指数开销的情况下实现了零方差估计 - 关于现实世界复杂网络（互联网）上量子空间搜索的硕士学位论文——以 PRResearch 论文形式发表 - 隶属于 QTF 卓越中心和 InstituteQ

30. Aaron Fitzpatrick

职位: Researcher (Junior Researcher) LinkedIn: https://www.linkedin.com/in/aaron-fitzpatrick-8004931bb/ Google Scholar: Profile exists at Semantic Scholar (ID: 2065154026) but URL could not be fetched

学历: PhD-level researcher (formal degree not publicly confirmed) —Quantum Chemistry / Computational Chemistry (Not publicly available) 毕业院校: Trinity Quantum Alliance, Trinity College Dublin

职业经历: - Algorithmiq Ltd: Junior Researcher (2022 - present) - Trinity Quantum Alliance, TCD: Researcher (Not specified)

研究方向: 变分量子本征求解器（VQE）方法, 轨道优化技术, 用于量子-经典方法的可极化嵌入（PE）, 近期量子器件的动态电子关联, 从量子测量重构密度矩阵

在 Algorithmiq 的角色: 初级研究员——从事量子化学的混合量子-经典算法研究，特别是具有轨道优化和可极化嵌入的 VQE 方法 h-index: 3-5 职业阶段: 早期职业研究者——在高影响力量子化学期刊上发表过多篇同行评审论文

代表论文: - Self-Consistent Field Approach for the Variational Quantum Eigensolver: Orbital Optimization Goes Adaptive (J. Phys. Chem. A, 2023) —first author - The variational quantum eigensolver self-consistent field method within a polarizable embedded framework (J. Chem. Phys. 160, 124114, 2024) —co-author - Electric Field Gradient Calculations for Ice VIII and IX Using Polarizable Embedding (J. Phys. Chem. A, 2024) —co-author - Quantum-centric strong and dynamical electron correlation: resource-efficient NEVPT2 for near-term quantum devices (arXiv:2405.15422, 2024) —co-author

主要成就: - 开发了用于轨道优化的 ADAPT-VQE-SCF 方法 - 为结合可极化嵌入与量子计算的 PE-VQE-SCF 方法做出了贡献 - 邮箱: aaron.fitzpatrick@algorithmiq.fi; ORCID: 0009-0007-9921-3522

31. Aaron Miller

职位: Researcher (Junior Researcher) LinkedIn: https://www.linkedin.com/in/aaron-miller-07a990197/ Google Scholar: Not publicly found

学历: PhD student-level —Physics / Quantum Information (Not publicly available) 毕业院校: School of Physics, Trinity College Dublin, Trinity Quantum Alliance, Dublin

职业经历: - Algorithmiq Ltd: Junior Researcher (2022-present) - School of Physics, Trinity College Dublin: PhD Researcher (Not specified) - Trinity Quantum Alliance: Researcher (Not specified)

研究方向: 费米子到量子比特映射（Bonsai、Treespilation 算法）, 量子线路编译, 变分量子本征求解器（VQE）, 核-电子轨道（NEO）方法, 量子处理器上的分子模拟

在 Algorithmiq 的角色: 初级研究员—主导费米子到量子比特映射（Bonsai, Treespilation）及分子模拟量子线路编译工作；同时隶属于 Trinity College Dublin h-index: 4-5 职业阶段: 职业生涯早期阶段研究员，拥有强劲的发表记录—在 Trinity College Dublin 从事博士水平的工作，在 PRX Quantum 和 arXiv 上发表了有影响力的第一作者论文

代表论文: - Bonsai Algorithm: Grow Your Own Fermion-to-Qubit Mappings (PRX Quantum 4, 030314, 2023) —first author - Treespilation: Architecture- and State-Optimised Fermion-to-Qubit Mappings (arXiv:2403.03992, 2024) —first author - Toward Accurate Post-Born-Oppenheimer Molecular Simulations on Quantum Computers (J. Chem. Theory Comput., 2023) —co-author - Approximate quantum circuit compilation for proton-transfer kinetics on quantum processors (Phys. Chem. Chem. Phys. 28, 3035, 2026) —co-author (AstraZeneca, IBM Quantum collaboration)

主要成就: - 开发了用于自动生成优化费米子到量子比特映射的 Bonsai 算法 - 开发了 Treespilation，使 ADAPT-VQE 的 CNOT 门计数减少高达 74% - 与 AstraZeneca、IBM Quantum、University of Cambridge 合作进行质子转移动力学模拟 - ORCID: 0000-0001-7108-1811; Email: aaron.miller@algorithmiq.fi

32. Leander Thiessen

职位: Researcher (Junior Researcher) LinkedIn: https://www.linkedin.com/in/leander-thiessen-9285a0113/ Google Scholar: Not publicly found (INSPIRE author ID: L.Thiessen.1 at https://inspirehep.net/authors/29

学历: Not publicly available —Quantum Chemistry (Not publicly available) 毕业院校: Affiliated with Algorithmiq Ltd, Helsinki, Finland

职业经历: - Algorithmiq Ltd: Junior Researcher (2024-present)

研究方向: 激发态量子化学, ADAPT-VQE 方法, 用于光动力疗法（PDT）的 BODIPY 光敏剂

在 Algorithmiq 的角色：Algorithmiq 公司青年研究员——利用 VQE 方法进行激发态分子模拟的量子化学算法研究 h-index：1 职业阶段：职业生涯非常早期——作为合著者发表了一篇同行评审论文；网络痕迹极少

代表论文：- Toward Accurate Calculation of Excitation Energies on Quantum Computers with ΔADAPT-VQE: A Case Study of BODIPY Derivatives (J. Phys. Chem. Lett. 15, 7111-7117, 2024; arXiv:2404.16149) —co-author (citations: 5 on INSPIRE)

主要成就：- 合著发表了关于 BODIPY 光敏剂的量子计算方法的论文，激发能的平均绝对误差低至 0.054 eV——性能优于经典的 TDDFT 和 EOM-CCSD 方法

33. Hetta Vappula

职位： Junior Researcher LinkedIn： https://www.linkedin.com/in/hetta-vappula-628055149/ Google Scholar: Not publicly found (Semantic Scholar author: https://www.semanticscholar.org/author/Hetta-Vappula/2319602989)

学历：Not publicly confirmed (likely BSc/MSc-level) —Quantum Information / Physics (Not publicly available) 毕业院校：University of Helsinki (affiliation inferred from collaborations), Algorithmiq Ltd, Helsinki

职业经历：- Algorithmiq Ltd: Junior Researcher (2024-present)

研究方向：近期量子硬件上的高精度测量，局部最优对偶框架的阴影估计，读出误差缓解，IBM 硬件上的量子化学应用

在 Algorithmiq 的角色：初级研究员——致力于近期量子设备的实用误差缓解和高精度测量技术 h-index：1-2 职业阶段：职业起步阶段——两篇合著论文（其中一篇发表于《npj Quantum Information》）；可能刚毕业或处于硕士/博士早期阶段

代表论文：- Practical Techniques for High-Precision Measurements on Near-Term Quantum Hardware: A Case Study in Molecular Energy Estimation (npj Quantum Information 11, 110, 2025; arXiv:2409.02575) —co-author - Improving shadow estimation with locally-optimal dual frames (arXiv:2511.02555, 2025) —co-author

主要成就：- 合著《npj Quantum Information》论文，展示了在 IBM Eagle r3 量子处理器上将测量误差从 1-5% 降低至 0.16% - 对局部最优阴影估计技术做出贡献

34. Francesco Grieco

职位： Junior Researcher LinkedIn： https://www.linkedin.com/in/francesco-grieco-58212a346/ Google Scholar: Not found

学历：Not publicly available —Not publicly identified (Not publicly available) 毕业院校：Algorithmiq Ltd, Helsinki

职业经历：- Algorithmiq Ltd: Junior Researcher (2024-present)

研究方向：未公开确认——未发现任何出版物

在 Algorithmiq 的角色：初级研究员——具体研究重点未公开记录 h-index：0 职业阶段：入门级——尚未发现公开出版物、学术背景或显著成就。可能系近期毕业生（从名字推测可能来自意大利），于 2024 年或之后作为初级团队成员加入 Algorithmiq。注意：此为 Algorithmiq 的 Francesco Grieco；根特大学另有一位从事天体物理学的 Francesco Grieco（ORCID: 0000-0003-0726-2348），并非同一人。

代表论文：- No publications found in public databases or search engines

主要成就：- 在 Algorithmiq 官方团队页面列为初级研究员

35. Fabio Tarocco

职位：Researcher LinkedIn: https://www.linkedin.com/in/quantumfabiotarocco/ Google Scholar: Not publicly indexed; AD Scientific Index profile available at https://adscientificindex.com/scientist/fabio-tarocco/5178248

学历: PhD Student (in progress) —Quantum Computation (Ongoing) 毕业院校: University of L’Aquila, University of Verona

职业经历: - University of Verona: Master’s Student in Engineering and Computer Science (2020-2022)

研究方向: 变分量子特征求解器（VQE）, 量子近似优化算法（QAOA）, 量子退火, 量子游走与图问题, 量子化学 - 电子结构与活性空间选择

在 Algorithmiq 的角色: 研究员 - 致力于量子化学与量子计算研究，从事活性空间选择工作流和 VQE 算法工作 h-index: 2 (18 total citations) 职业阶段: 职业生涯早期（博士生/初级研究员）

代表论文: - AEGISS - Atomic orbital and Entropy-based Guided Inference for Space Selection (arXiv:2508.10671, 2025) - Scalable quantum circuit generation for iterative ground state approximation using Majorana Propagation (arXiv:2603.23444, 2026) - Computing graph edit distance on quantum devices (with M. Incudini and A. Mandarino, Springer Nature)

主要成就: - 作为 Algorithmiq 团队成员，因量子驱动的癌症药物发现而获得 200 万美元 Wellcome Leap Q4Bio 奖（2026 年）- 获得 PNRR 资助，在 University of L’Aquila 攻读量子计算博士项目

36. Francesca Pietracaprina

职位: Performance Engineer LinkedIn: https://www.linkedin.com/in/francesca-pietracaprina/ Google Scholar: Not directly found; indexed on zbMATH (https://zbmath.org/authors/pietracaprina.francesca), MathSciNet (Author ID 1220640), and arXivLens (https://arxivlens.com/Author/Details/francesca-pietracaprina-2141-25463508)

学历: PhD —Statistical Physics / Quantum Many-Body Physics (2015) 毕业院校: SISSA (Trieste), Sapienza University of Rome, University of Toulouse (LPT, CNRS), Trinity College Dublin

职业经历: - Sapienza University of Rome: Postdoctoral Researcher (2015-2017) - University of Toulouse (LPT, CNRS): Postdoctoral Researcher (2017-2019) - Trinity College Dublin (QUSYS Group): Marie Sklodowska-Curie Fellow (2019-2021) - Algorithmiq: Software Engineer / Performance Engineer (2022-present)

研究方向: 无序量子系统中的多体局域化, Hilbert 空间碎裂与多重分形, 多体系统中的量子纠缠, Bethe 晶格上的 Anderson 局域化, 量子热力学

在 Algorithmiq 的角色: 性能工程师（亦列为软件工程师与研究员）——专注于优化量子算法性能及全栈量子计算解决方案，以应用于药物发现与模拟领域 h-index: 11 (21-23 publications, ~473 total citations) 职业阶段: 职业中期（博士 + 多项博士后经历 + 玛丽·居里奖学金 + 超过 3 年的行业经验）

代表论文: - Hilbert-space fragmentation, multifractality, and many-body localization - Annals of Physics (2021) - Probing many-body localization in a disordered quantum dimer model on the honeycomb lattice - SciPost Phys. 10, 044 (2021) - Logarithmic growth of local entropy and total correlations in many-body localized dynamics - Quantum (2020, with J. Goold) - Anderson transition on the Bethe lattice: an approach with real energies - J. Stat. Mech. (2020)

主要成就: - 都柏林圣三一学院玛丽·居里学者（欧盟 Horizon 2020 计划）- 参与获得欧洲研究委员会概念验证资助的研究项目，将统计物理学工具应用于卫星图像分析以评估荒漠化风险 - 2024 年林道诺贝尔奖得主大会参会者 - 接受 DECODE Quantum 播客专访（2024 年第 76 期）- 与诺贝尔奖得主 Giorgio Parisi 合作

37. Ramon Panades Barrueta

职位: Software Engineer LinkedIn: https://www.linkedin.com/in/rpanades/ Google Scholar: https://scholar.google.co

学历: PhD in Physics —Quantum Chemistry / Theoretical Physics (2020) 毕业院校: Universite de Lille, University of Havana

职业经历: - TU Dresden (NOMAD Laboratory): Postdoctoral Researcher (2021-2023) - University of Twente (TREX Project): Postdoctoral Researcher (2020-2021) - Heidelberg University: Visiting PhD Student (2019) - University of Lille 1: Teaching Assistant (2018-2019)

研究方向: 量子计算与张量网络算法（混合量子-经典）, 在 FHI-aims 和 GreenX 中实现的用于核心级光谱学（XPS/XAS）的低标度 GW 基方法, 大规模并行量子蒙特卡洛算法（CHAMP 代码，TREX CoE）, 核量子动力学与张量分解（MCTDH 方法）, 势能面生成（SRP-MGPF、SOP-FBR、CP-FBR 方法）

在 Algorithmiq 的角色: 软件工程师 - 使用张量网络方法实现前沿的高性能混合量子-经典算法；致力于编码最佳实践和稳健的软件设计 h-index: 1-2 (5 publications, ~16 citations) 职业阶段: 早期到中期职业（博士+3 年博士后 + 产业界职位）

代表论文: - Accelerating core-level GW calculations by combining the contour deformation approach with the analytic continuation of W - J. Chem. Theory Comput. (2023, with Golze) - Double excitation energies from quantum Monte Carlo using state-specific energy optimization - J. Chem. Theory Comput. (2022, with Shepard et al.) - Low-Rank Sum-of-Products Finite-Basis-Representation (SOP-FBR) of Potential Energy Surfaces - J. Chem. Phys. (2020) - Specific Reaction Parameter Multigrid POTFIT (SRP-MGPF): Automatic Generation of Sum-of-Products Form Potential Energy Surfaces for Quantum Dynamical Calculations - Frontiers in Chemistry, 7:576 (2019)

主要成就: - 博士论文：大气分子与模型烟尘颗粒相互作用的量子模拟（Universite de Lille）- 用于自动化势能面生成的 SRPTucker 软件包的开发者 - 开源量子化学代码（FHI-aims、GreenX、CHAMP）的贡献者 - ORCID: 0000-0003-4239-0978

38. Cannur Floszmann

职位: Junior e-Learning Developer LinkedIn: https://www.linkedin.com/in/cannur-floszmann/ Google Scholar: Not found

学历: Not publicly available —Not publicly available (Not publicly available)

在 Algorithmiq 的角色: 初级电子学习开发员——负责开发电子学习内容和教育材料（可能与量子计算教育平台相关）h-index: N/A 职业阶段: 早期职业（初级职位）

39. Elsi-Mari Borrelli

职位: Country Manager, Finland LinkedIn: https://www.linkedin.com/in/elsi-mari-borrelli/ Google Scholar: https://scholar.google.com/citations?user=J9G1CR8AAAAJ

学历: PhD in Theoretical Physics —Theoretical Physics / Quantum Physics (Not publicly specified) 毕业院校: University of Helsinki (presumed), Aalto University (postdoc), University of Turku (postdoc)

职业经历: - ABB Corporate Research Switzerland: Senior Scientist (~2018-2022) - Aalto University: Postdoctoral Researcher () - University of Turku: Postdoctoral Researcher ()

研究方向: 量子生物学与量子网络医学, 光动力疗法（PDT）光敏剂设计, 量子错误缓解与测量断层成像, 非马尔可夫开放量子系统, ADAPT-VQE 变分量子算法

在 Algorithmiq 的角色: 芬兰国家经理（同时在 Algorithmiq 团队页面上列为企业合作与销售主管；曾任首席量子生物学研究员）h-index: ~15 (estimated based on highly-cited papers in non-Markovian dynamics) 职业阶段: 从学术和工业研发转向商业领导的中期职业研究员/经理

代表论文: - Colloquium: Non-Markovian dynamics in open quantum systems - Reviews of Modern Physics (2016) - 800+ citations - Measure for the degree of non-Markovian behavior - Physical Review Letters (2009) - 1,300+ citations - Toward Accurate Calculation of Excitation Energies on Quantum Computers with Delta-ADAPT-VQE: A Case Study of BODIPY Derivatives - J. Phys. Chem. Lett. (2024) - Quantum Computing for Photosensitizer Design in Photodynamic Therapy - Annual Review of Biomedical Data Science (2025)

主要成就: - 合著了《Reviews of Modern Physics》中关于非马尔可夫动力学的开创性座谈会论文 - 量子网络医学新兴领域的先驱 - Algorithmiq 量子生物学和光动力疗法研究项目的主要贡献者 - InstituteQ - 芬兰量子研究所成员

40. David Lin

职位: General Counsel LinkedIn: https://www.linkedin.com/in/dklin/ Google Scholar: N/A

学历: Juris Doctor (J.D.) —Electrical Engineering / Law (Not publicly specified) 毕业院校: UCLA - B.S. in Electrical Engineering, Loyola Law School, Los Angeles - J.D.

职业经历: - Winston & Strawn LLP: BigLaw Litigator (2013-2023) - davidlin.law, PC: Founder () - DTO Law (Los Angeles): Counsel (Current)

在 Algorithmiq 的角色: 法务总监——负责知识产权战略、技术商业交易和合作协议。同时担任 DTO Law 的法律顾问。常驻加利福尼亚州帕萨迪纳。h-index: N/A 职业阶段: 资深法律专业人士，拥有 15 年以上在大型律师事务所从事专利诉讼和内部领导职位的经验

主要成就: - 代表三家主要视频游戏发行商在 PTAB 前成功无效了一个专利组合 - 在平行的地区法院诉讼中，通过简易判决成功击败了六项被主张的专利 - 在国际贸易委员会 (ITC) 审理了一桩涉及多项专利的案件，经历了完整的庭审过程 - 在六项以上的专利侵权诉讼中为一家领先的网络/通信公司进行辩护 - 被列为《The Quantum Law Navigator》(Chicago Quantum Exchange, 2025) 的撰稿人

41. Jason Gumarang

职位: IT Specialist LinkedIn: https://www.linkedin.com/in/jason-gumarang/ Google Scholar: N/A

学历: Not publicly available —Not publicly available (Not publicly available)

职业经历: - Not publicly available: Not publicly available (Not publicly available)

在 Algorithmiq 的角色: IT 专员 - 负责 IT 基础设施和系统支持 h-index: N/A 职业阶段: 未确定（公开信息极少 - 除 Algorithmiq 团队页面外无搜索结果）

主要成就: - 在 Algorithmiq 官方团队页面列为 IT 专员

四、产品线

Aurora

- 类型: 平台
- 状态: 商用
- 上线: 2024
- 描述: 量子化学分子模拟平台。用于药物发现、材料科学、光动力癌症疗法模拟。

TEM (Tensor-network Error Mitigation)

- 类型: 工具
- 状态: 商用
- 上线: 2024-09
- 描述: 基于张量网络的后处理误差缓解。91 量子比特 x4095 门验证。IBM Qiskit Catalog 销售。

Digital Quantum Interface (DQI)

- 类型: 平台
- 状态: 商用
- 上线: 2024
- 描述: 连接量子计算机与经典 HPC，专利信息完备测量 (IC-POVM)。

Measurement Toolbox

- 类型: 工具
- 状态: 商用
- 上线: 2024
- 描述: 信息完备测量优化，减少误差并节约计算资源。

Circuit Optimisation Toolbox

- 类型：工具
- 状态：商用
- 上线：2024
- 描述：最小化量子电路中的门数和操作，降低噪声。

五、合作伙伴

合作方	日期	类型	描述

六、技术里程碑

日期	事件	意义

七、SWOT 分析

优势

劣势

机会

威胁

八、完整论文列表（70 篇）

#	日期	类型	标题	作者	期刊	主题
1	2026-03	预印本	Scalable quantum circuit generation for iterative ground state approximation using Majorana Propagat	Rahul Chakraborty, Aaron Miller, Anton Nykänen, Özlem Salehi, Fabio Tarocco, Fab	arXiv	算法，模拟
2	2026-02	预印本	Experimental implementation of a discrete-time quantum walk on biological networks	Viacheslav Dubovitskii, Filippo Utro, Aritra Bose, Laxmi Parida, Sabrina Manisca	arXiv	生物，量子信息

#	日期	类型	标题	作者	期刊	主题
3	2025-11	预印本	Link prediction with swarms of chiral quantum walks	Gaia Forghieri, Viacheslav Dubovitskii, Matteo A. C. Rossi, Matteo G. A. Paris	arXiv	量子信息
4	2025-11	预印本	Fermionic Born Machines: Classical training of quantum generative models based on Fermion Sampling	Bence Bakó, Zoltán Kolarovszki, Zoltán Zimborás	arXiv	量子信息, ML
5	2025-11	预印本	Improving shadow estimation with locally-optimal dual frames	Keijo Korhonen, Stefano Mangini, Joonas Malmi, Hetta Vappula, Daniel Cavalcanti	arXiv	量子信息
6	2025-08	预印本	AEGISS - Atomic orbital and Entropy-based Guided Inference for Space Selection -A novel semi-auto	Fabio Tarocco, Pi A. B. Haase, Fabijan Pavoševi , Vijay Krishna, Leonardo Guidon	arXiv	化学
7	2025-07	预印本	Approximate quantum circuit compilation for proton-transfer kinetics on quantum processors	Arseny Kovyrshin et al.	arXiv	化学, 算法
8	2025-03	预印本	Simulation of Fermionic circuits using Majorana Propagation	Aaron Miller, Joachim Favre, Zoë Holmes, Özlem Salehi, Rahul Chakraborty, Anton	arXiv	算法, 模拟, 量子信息

#	日期	类型	标题	作者	期刊	主题
9	2025-01	预印本	Myths around quantum computation before full fault tolerance: What no-go theorems rule out and what	Zoltán Zimborás et al.	arXiv	
10	2024-12	预印本	A generic multi-Pauli compilation framework for limited connectivity	Adam Glos, Özlem Salehi	arXiv	算法
11	2024-10	预印本	The advent of fully variational quantum eigensolvers using a hybrid multiresolution approach	Fabian Langkabel, Stefan Knecht, Jakob S. Kottmann	arXiv	
12	2024-05	预印本	Quantum-centric strong and dynamical electron correlation: A resource-efficient second-order N-ele	Aaron Fitzpatrick, N. Walter Talarico, Roberto Di Remigio Eikås, Stefan Knecht	arXiv	模拟
13	2024-05	预印本	Problem-informed Graphical Quantum Generative Learning	Bence Bakó, Dániel T. R. Nagy, Péter Hága, Zsófia Kallus, Zoltán Zimborás	arXiv	ML
14	2024-05	预印本	Generalized group designs: overcoming the 4-design-barrier and constructing novel unitary 2-designs	Ágoston Kaposi, Zoltán Kolarovszki, Adrián Solymos, Zoltán Zimborás	arXiv	

#	日期	类型	标题	作者	期刊	主题
15	2024-04	预印本	ΔADAPT-VQE: Toward Accurate Calculation of Excitation Energies on Quantum Computers for BODIPY Molec	Anton Nykänen, Leander Thiessen, Elsi-Mari Borrelli, Vijay Krishna, Stefan Knech	arXiv	生物
16	2024-03	预印本	Scalability of quantum error mitigation techniques: from utility to advantage	Sergey N. Filippov, Sabrina Maniscalco, Guillermo García-Pérez	arXiv	误差缓解
17	2024-03	预印本	Treespilation: Architecture- and State-Optimised Fermion-to-Qubit Mappings	Aaron Miller, Adam Glos, Zoltán Zimborás	arXiv	
18	2023-11	预印本	Disease Gene Prioritiza-tion With Quantum Walks	Harto Saarinen, Mark Goldsmith, Rui-Sheng Wang, Joseph Loscalzo, Sabrina Manisca	arXiv	量子信息
19	2023-10	预印本	Variational quantum eigensolver boosted by adiabatic connection	Mikuláš Matoušek, Katarzyna Pernal, Fabijan Pavošević, Libor Veis	arXiv	
20	2023-10	预印本	Toward Accurate Post-Born-Oppenheimer Molecular Simulations on Quantum Computers: An Adaptive Variat	Anton Nykänen et al.	arXiv	化学, 模拟

#	日期	类型	标题	作者	期刊	主题
21	2023-09	预印本	Unitary Coupled-Cluster for Quantum Computation of Molecular Properties in a Strong Magnetic Field	Tanner Culpitt, Erik I. Tellgren, Fabijan Pavosevic	arXiv	化学
22	2023-09	预印本	High performance Boson Sampling simulation via data-flow engines	Gregory Morse, Tomasz Rybotycki, Ágoston Kaposi, Zoltán Kolarovszki, Uroš Stoj i	arXiv	模拟
23	2023-07	预印本	Scalable tensor-network error mitigation for near-term quantum computing	Sergei Filippov, Matea Leahy, Matteo A. C. Rossi, Guillermo García-Pérez	arXiv	误差缓解
24	2023-07	预印本	Towards quantum-enabled cell-centric therapeutics	Saugata Basu et al.	arXiv	
25	2023-02	预印本	Efficient qudit based scheme for photonic quantum computing	Márton Karácsony, László Oroszlány, Zoltán Zimborás	arXiv	
26	2022-12	预印本	Matrix product channel: Variationally optimized quantum tensor network to mitigate noise and reduce	Sergey Filippov, Boris Sokolov, Matteo A. C. Rossi, Joonas Malmi, Elsi-Mari Borr	arXiv	误差缓解，张量网络

#	日期	类型	标题	作者	期刊	主题
27	2022-08	预印本	Adaptive POVM implementations and measurement error mitigation strategies for near-term quantum devi	Adam Glos, Anton Nykänen, Elsi-Mari Borrelli, Sabrina Maniscalco, Matteo A. C. R	arXiv	误差缓解
28	2022-07	预印本	Quantifying Electron Entanglement Faithfully	Lexin Ding, Zoltan Zimboras, Christian Schilling	arXiv	量子信息
29	2022-07	预印本	Virtual linear map algorithm for classical boost in near-term quantum computing	Guillermo García-Pérez, Elsi-Mari Borrelli, Matea Leahy, Joonas Malmi, Sabrina M	arXiv	算法
30	2022-06	预印本	Quantum network medicine: rethinking medicine with network science and quantum algorithms	Sabrina Maniscalco et al.	arXiv	算法
31		期刊	Error-Mitigation Enabled Multicomponent Quantum Simulations Beyond the Born-Oppenheimer Approximatio	Delmar G. A Cabral, Brandon Allen, Fabijan Pavošević, Sharon Hammes-Schiffer, Pa	ACS	化学，模拟
32		期刊	Nuclear-electronic orbital second-order coupled cluster for excited states	Jonathan H. Fetherolf, Fabijan Pavošević, Sharon Hammes-Schiffer	J. Chem. Phys.	化学
33		期刊	Dynamical simulations of many-body quantum chaos on a quantum computer	Laurin E. Fischer et al.	Nature	模拟

#	日期	类型	标题	作者	期刊	主题
34		期刊	Origin of the insulating phase and metal-insulator transition in the organic molecular solid κ-(BEDT	Dongbin Shin, Fabijan Pavošević, Nicolas Tancogne-Dejean, Michele Buzzi, Emil Vi	Nature	化学
35		期刊	Approximate quantum circuit compilation for proton-transfer kinetics on quantum processors	Arseny Kovyrshin et al.	RSC	化学，算法
36		期刊	Constrained Nuclear-Electronic Orbital Theory for Quantum Computation	Tanner Culpitt, Zenhua Chen, Fabijan Pavošević, Yang Yang	ACS	化学
37		期刊	Low variance estimations of many observables with tensor networks and informationally-complete measu	Stefano Mangini, Daniel Cavalcanti	Quantum	张量网络
38		期刊	Practical techniques for high-precision measurements on near-term quantum hardware and applications	Keijo Korhonen, Hetta Vappula, Adam Glos, Marco Cattaneo, Zoltán Zimborás, Elsi-	Nature	化学
39		期刊	Small tensor product distributed active space (STP-DAS) framework for relativistic and non-relativis	Hang Hu, Shiv Upadhyay, Lixin Lu, Andrew J. Jenkins, Tianyuan Zhang, Agam Shayit	J. Chem. Phys.	张量网络

#	日期	类型	标题	作者	期刊	主题
40		期刊	Estimating molecular thermal averages with the quantum equation of motion and information-ally comple	Daniele Morrone, N. Walter Talarico, Marco Cattaneo, Matteo A. C. Rossi	MDPI	化学
41		期刊	Enhanced observable estimation through classical optimization of informa-tionally overcomplete measur	Joonas Malmi, Keijo Korhonen, Daniel Cavalcanti, Guillermo García-Pérez	Physical Review	
42		期刊	Efficient and operational quantifier of non-divisibility in terms of channel dis-crimination	Ranieri Nery, Nadja K. Bernardes, Daniel Cavalcanti, Rafael Chaves, Cristhiano D	Physical Review	
43		期刊	Self-Consistent Field Approach for the Variational Quantum Eigensolver: Orbital Optimization Goes Ad	Aaron Fitzpatrick, Anton Nykänen, N. Walter Talarico, Alessandro Lunghi, Sabrina	ACS	化学
44		期刊	Transfer and routing of Gaussian states through quantum complex networks with and without community	Markku Hahto, Johannes Nokkala, Guillermo García-Pérez, Sabrina Maniscalco, Jyrk	Quantum	
45		期刊	Piquasso: A Photonic Quantum Computer Simulation Software Platform	Zoltán Kolarovszki et al.	Quantum	模拟

#	日期	类型	标题	作者	期刊	主题
46		期刊	Towards Efficient Quantum Computing for Quantum Chemistry: Reducing Circuit Complexity with Transcor	Erika Magnusson, Aaron Fitzpatrick, Stefan Knecht, Martin Rahm, Werner Dobrautz	RSC	化学，算法
47		期刊	Hamiltonian-oriented homotopy quantum approximate optimization algorithm	Akash Kundu, Ludmila Botelho, Adam Glos	Physical Review	算法
48		期刊	Tensor network noise characteriza-tion for near-term quantum computers	Stefano Mangini, Marco Cattaneo, Daniel Cavalcanti, Sergei Filippov, Matteo A. C	Physical Review	误差缓解，张量网络
49		期刊	Batched Line Search Strategy for Navigating through Barren Plateaus in Quantum Circuit Training	Jakab Nádori, Gregory Morse, Barna Fülöp Villám, Zita Majnay-Takács, Zoltán Zimb	Quantum	算法
50		期刊	Calculating the many-body density of states on a digital quantum computer	Alessandro Summer, Cecilia Chiaracane, Mark T. Mitchison, John Goold	Physical Review	模拟
51		期刊	Quantum Many-Body Scars in Dual-Unitary Circuits	Leonard Logari , Shane Dooley, Silvia Pappalardi, John Goold	Physical Review	算法，模拟

#	日期	类型	标题	作者	期刊	主题
52		期刊	Classification and transformations of quantum circuit decompositions for permutation operations	Ankit Khandelwal, Handy Kurniawan, Shraddha Aangiras, Özlem Salehi, Adam Glos	Springer	算法
53		期刊	The variational quantum eigensolver self-consistent field method within a polarizable embedded frame	Erik Rosendahl Kjellgren, Peter Reinholdt, Aaron Fitzpatrick, Walter N. Talarico	J. Chem. Phys.	
54		期刊	Simulating polaritonic ground states on noisy quantum devices	Mohammad Hassan, Fabijan Pavoševi , Derek S. Wang, Johannes Flick	ACS	模拟
55		期刊	Self-consistent quantum measurement tomography based on semidefinite programming	Marco Cattaneo, Matteo A. C. Rossi, Keijo Korhonen, Elsi-Mari Borrelli, Guillerm	Physical Review	
56		期刊	Spin-Flip Unitary Coupled Cluster Method: Toward Accurate Description of Strong Electron Correlation	Fabijan Pavoševi , Ivano Tavernelli, Angel Rubio	ACS	

#	日期	类型	标题	作者	期刊	主题
57		期刊	Bonsai Algorithm: Grow Your Own Fermion-to-Qubit Mappings	Aaron Miller, Zoltán Zimborás, Stefan Knecht, Sabrina Maniscalco, Guillermo Garc	Physical Review	算法
58		期刊	Evidence of Kardar-Parisi-Zhang scaling on a digital quantum simulator	Nathan Keenan, Niall F. Robertson, Tara Murphy, Sergiy Zhuk, John Goold	Nature	
59		期刊	Long-time relaxation of a finite spin bath linearly coupled to a qubit	Jukka P. Pekola, Bayan Karimi, Marco Cattaneo, Sabrina Maniscalco	World Scientific	
60		期刊	Experimentally accessible nonseparabil-ity criteria for multipartite-entanglement-structure detection	Guillermo García-Pérez, Oskari Kerppo, Matteo A. C. Rossi, Sabrina Maniscalco	Physical Review	量子信息
61		期刊	Variational Gibbs State Preparation on NISQ devices	Mirko Consiglio, Jacopo Settino, Andrea Giordano, Carlo Mastroianni, Francesco P	Physical Review	
62		期刊	Quantum Simulation of Dissipative Collective Effects on Noisy Quantum Computers	Marco Cattaneo, Matteo A.C. Rossi, Guillermo García-Pérez, Roberta Zambrini, Sab	Physical Review	模拟
63		期刊	Local quantum overlapping tomography	Bruna G. M. Araújo, Márcio M. Taddei, Daniel Cavalcanti, Antonio Acín	Physical Review	

#	日期	类型	标题	作者	期刊	主题
64		期刊	Mitigating the measurement overhead of ADAPT-VQE with optimised informationally complete generalised	Anton Nykänen, Matteo A. C. Rossi, Elsi-Mari Borrelli, Sabrina Maniscalco, Guill	Physical Review	
65		期刊	Spatial search by continuous-time quantum walks on renormalized Internet networks	Joonas Malmi, Matteo A. C. Rossi, Guillermo García-Pérez, Sabrina Maniscalco	Physical Review	量子信息
66		期刊	Steering-based randomness certification with squeezed states and homodyne measurements	Marie Ioannou, Bradley Longstaff, Mikkel V. Larsen, Jonas S. Neergaard-Nielsen,	Physical Review	
67		期刊	Prog-QAOA: Framework for resource-efficient quantum optimization through classical programs	Bence Bakó, Adam Glos, Özlem Salehi, Zoltán Zimborás	Quantum	
68		期刊	Relativistic Kramers-Unrestricted Exact-Two-Component Density Matrix Renormalization Group	Chad E. Hoyer, Hang Hu, Lixin Lu, Stefan Knecht, Xiaosong Li	ACS	
69		期刊	Combining the in-medium similarity renormalization group with the density matrix renormalization gro	A. Tichai, S. Knecht, A. T. Kruppa, Ö. Legeza, C. P. Moca, A. Schwenk, M. A. Wer	Elsevier	

#	日期	类型	标题	作者	期刊	主题
70		期刊	Learning to Measure: Adaptive Informationally Complete Generalized Measurements for Quantum Algorith	Guillermo García-Pérez, Matteo A.C. Rossi, Boris Sokolov, Francesco Tacchino, Pa	Physical Review	算法, ML

九、交叉分析

时间线

年份	融资	产品	论文
2021	种子轮 $4M	—	起步
2022	—	—	5 篇
2023	A 轮 $15M	—	8 篇
2024	—	5 款产品发布	8 篇
2025	Q4Bio $2M	TEM 商用	7 篇
2026	B 轮 €18M	—	2 篇（进行中）

论文主题分布

- 算法 (quantum_algorithms): 14 篇
- 化学 (quantum_chemistry): 13 篇
- 模拟 (quantum_simulation): 12 篇
- 量子信息 (quantum_information): 9 篇
- 误差缓解 (error_mitigation): 5 篇
- 张量网络 (tensor_networks): 4 篇
- ML (machine_learning): 3 篇
- 生物 (quantum_biology): 2 篇

年度产出

- 2022: 5 篇
- 2023: 8 篇
- 2024: 8 篇
- 2025: 7 篇
- 2026: 2 篇

最后更新: ?